KSHITIJ ASHOK SAWARKAR

Autonomous Robotics, Embedded Systems & Forward Deployed Engineering

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Executive Summary

Autonomous systems and forward-deployed automation engineer with an M.Tech in Control and Instrumentation from MNNIT Allahabad and a B.Tech in Electronics and Communication Engineering from RCOEM Nagpur. Operates at the intersection of embedded hardware, deterministic edge runtimes, real-time computer vision, and industrial systems automation. Lead author of peer-reviewed edge-perception research published in IEEE Xplore (Document ID: 10627276). Proven track record of delivering production-grade client-side engines executing under 50ms latency, software-defined radio (SDR) spectral sensing pipelines for UAV RF classification, and ROS2 embedded robotics control stacks.

IEEE: Smart Shopping Cart | IEEE Conference Publication | IEEE Xplore

Core Technical Competencies

Domain	Key Technologies & Methodologies
Autonomous Robotics & Embedded	ROS2 (composition nodes, lifecycle management), Embedded Linux, Microcontrollers (Raspberry Pi, Arduino), Serial Protocols (CAN, UART, SPI, I2C), Sensor Synchronization (LiDAR, Monocular Vision, IMU), Hardware-in-the-Loop (HIL).

Domain	Key Technologies & Methodologies
Applied AI & Edge Perception	Real-Time Computer Vision (YOLOv5, OpenCV), Model Quantization (INT8), Edge Compute Optimization, Spectral Feature Extraction, Machine Learning (TensorFlow, Keras, Scikit-Learn).
RF Telemetry & Signal Processing	Software-Defined Radio (SDR), RF Signal Classification, Spectrogram Analysis, IQ Signal Parsing, UAV Flight Controller Telemetry Interception.
Production Systems & Cloud	Cloudflare Pages, Deterministic Parsing Algorithms, Client-Side Runtime Optimization, AWS (EC2, Lambda, S3, SageMaker), RESTful APIs, Git, Docker, Shell Scripting.

Education & Academic Pedigree

Master of Technology (M.Tech) in Control and Instrumentation

Motilal Nehru National Institute of Technology (MNNIT) Allahabad | *2025 - Present*

- Specialization in high-order dynamic systems, state-space modeling, multivariable feedback control, and real-time embedded signal processing
- Serving as Teaching Assistant for undergraduate control systems lectures and laboratory tutorials.
- Advanced Coursework: NPTEL "Introduction to Multi-Agent Systems" (IIT Certification).

Bachelor of Technology (B.Tech) in Electronics and Communication Engineering

Shri Ramdeobaba College of Engineering and Management (RCOEM), Nagpur | *2020 - 2024*

- Core foundations in Microprocessors & Microcontrollers, Digital Signal Processing (DSP), RF & Microwave Systems, Embedded Hardware Design, and Communication Networks.

Professional Experience & Internships

Wakefit Innovations Ltd. | Bengaluru / Hosur, India

Systems Automation Intern - PGET | *August 2026 - Present*
Live:: NPL Automation | Wakefit MicroSaaS | Image Sequence Renamer |

- **Combinatorial Schema Engine:** Architected and deployed a production client-side application (`npl-automation.pages.dev`) on Cloudflare Pages, automating multi-tier Finished Goods (FG) code generation and New Product Launch (NPL) schema logic across complex seater variants and dynamic colorways.
- Process Latency Compression: Eliminated multi-day manual ERP spreadsheet reconciliation bottlenecks by implementing recursive client-side parsing algorithms, compressing end-to-end BOM derivation from 72+ operational hours into deterministic sub-minute runs with sub-50ms execution latency.
- Industrial Guardrails & Schema Isolation: Authored formal system constraints (`AGENTS.md`) and deterministic validation guardrails, preventing parameter collisions, duplicate SKU generation, and upstream ERP data ingestion errors across active manufacturing assembly lines.
- **Zero-Compute Infrastructure:** Engineered the entire schema pipeline to execute strictly client-side with 100% data isolation and zero server compute overhead, eliminating cloud infrastructure maintenance and ensuring uninterrupted offline plant-floor readiness.

AICTE Virtual Internship | Remote, India

AWS Cloud Intern / AI Developer | *May 2023 - July 2023*

- **Distributed Cloud Migration:** Architected and executed the migration of local machine learning workflows and structured datasets to AWS infrastructure, configuring EC2 instances, security groups, and IAM role policies for distributed execution.
- **Event-Driven Serverless Pipelines:** Designed asynchronous, serverless ETL data pipelines using AWS Lambda triggered by Amazon S3 bucket storage events, automating data ingestion, validation, and real-time preprocessing with zero idle compute costs.
- **Production Model Deployment & MLOps:** Managed full model training lifecycles and deployed hosted inference endpoints using AWS SageMaker, optimizing distributed algorithms (XGBoost, Linear Learner) for scalable real-time inference against production datasets.

- **Cloud Security & Scalability:** Implemented CloudWatch monitoring, automated alerts, and resource scaling policies to ensure continuous endpoint availability while preventing resource waste.

RCOEM TBI Foundation Cell | Nagpur, India

IoT Systems & Embedded Firmware Intern | *January 2021 - February 2022*

- **IoT Firmware & Hardware Interfacing:** Developed embedded firmware and sensor interfacing routines across microcontrollers and IoT development boards for connected hardware products incubated under Sharon Innovations.
- **Telemetry & Protocol Integration:** Integrated serial communication protocols (UART, SPI, I2C) to link environmental and operational sensors with edge communication modules, ensuring low-power telemetry transmission with zero packet loss.
- **Hardware-in-the-Loop Debugging:** Built internal test automation scripts and hardware debugging routines to profile voltage stability, sensor drift, and signal noise across early prototype iterations.
- **Prototype Stabilization for Pilot Deployments:** Resolved hardware-software integration bottlenecks across partner ventures, increasing overall prototype operational reliability prior to pilot field trials.

UCN Cable Networks | Nagpur, India

Computer Networking Intern | March 2023 - April 2023

- **Network Infrastructure & Routing Topology:** Configured and provisioned enterprise routing hardware, managed Layer 2 and Layer 3 (L2/L3) network switches, and established static and dynamic routing topologies across regional distribution points.
- **Subnet Architecture & Protocol Management:** Implemented IPv4 CIDR subnetting, VLAN segmentation, and DHCP/DNS server configurations to isolate operational traffic and optimize bandwidth allocation across mixed client networks.
- **Physical & Data Link Diagnostics:** Conducted physical Layer 1 and Layer 2 troubleshooting using packet-capture analyzers and cable-fault testers, identifying signal attenuation, packet drops, and MTU mismatch bottlenecks across physical trunk lines.
- **Network Reliability & Uptime Assurance:** Built and deployed automated network monitoring check-scripts to trace gateway latency, resolve broadcast storms, and maintain high-availability network operations for downstream distribution nodes.

Flagship Technical Projects & Production Deployments

Real-Time Drone Detection & RF Signal Classification

Python, SDR, RF Signal Processing, Machine Learning
Live::
https://docs.google.com/presentation/d/1lMEy7h-iVUmmVrr2GoM_qS5AJWuqM7LC51hbWQ7Y5Rg/edit?usp=sharing

- Built an automated edge spectral sensing pipeline intercepting communication bursts between commercial UAV flight controllers and Ground Control Stations (GCS) across 2.4 GHz and 5.8 GHz ISM bands.
- Configured Software-Defined Radio (SDR) hardware to stream raw I/Q signal data, applying Fast Fourier Transforms (FFT) and spectrogram processing to isolate low-power transmission spikes against ambient radio noise.
- Trained lightweight neural classification models to differentiate drone telemetry from Wi-Fi/Bluetooth interference with sub-second classification latency.

Manufacturing Schema & BOM Automation Engine (NPL Automation)

JavaScript, Cloudflare Pages, Deterministic Parsing
Live:: NPL Automation | Wakefit MicroSaaS

- Engineered an industrial client-side schema engine deployed on Cloudflare Pages, automating multi-tier Finished Goods (FG) code generation and New Product Launch (NPL) parameter trees across dynamic seater variants and colorways.
- Solved combinatorial explosion bottlenecks using recursive client-side parsing algorithms, compressing 72+ operational hours of manual ERP reconciliation into deterministic sub-minute execution with sub-50ms latency.
- Authored formal system constraints (AGENTS.md) and deterministic validation guardrails to prevent SKU collisions and upstream ERP data ingestion errors across active manufacturing assembly lines.
- Structured zero-compute infrastructure running 100% client-side with full data isolation and zero server overhead.

Smart Shopping Cart: Real-Time Edge Object Detection

Raspberry Pi, YOLOv5, OpenCV, Python
Live:: Smart Shopping Cart | IEEE Conference Publication | IEEE Xplore

- Curated a custom retail dataset and fine-tuned YOLOv5 models for multi-class edge detection running directly on ARM-constrained Raspberry Pi hardware.

- Implemented INT8 model parameter quantization and cache optimization to balance thermal throttling limits against frames-per-second (FPS) throughput.
- Designed an OpenCV pipeline executing local billing logic with zero server dependencies, guaranteeing full offline operational readiness.
- Published on IEEE Xplore.

HiWonder MentorPi M1 Autonomous Robotics Platform

Raspberry Pi 5, ROS2, Computer Vision, Robotics Telemetry
Images::
https://drive.google.com/file/d/1OSRXNmmUL0gwdEIcF-1FtI1Xcxfa-7rT/view?usp=sharing

- Programmed and tuned an autonomous mobile robot (AMR) unit powered by a Raspberry Pi 5 single-board computer running Robot Operating System 2 (ROS2).
- Configured modular ROS2 publisher/subscriber nodes for real-time motor actuation, PWM driver loops, and telemetry streaming on embedded Linux.
- Interfaced onboard monocular camera feeds with OpenCV pipelines for visual marker tracking, target acquisition, and spatial orientation.

TrendGuard-200: Multi-Asset Quantitative Risk Engine

Python, NumPy, Pandas, Vectorized Backtesting

- Architected a quantitative signal generation and backtesting engine in Python, evaluating multi-asset momentum regimes across an 8-year OHLC dataset.
- Accelerated computation cycles by 90% through vectorized NumPy and Pandas operations, achieving a 72.7% win rate.
- Structured strict drawdown-containment filters using dynamic position sizing and multi-timeframe moving average boundaries (21, 50, 200 SMA) alongside RSI and VWAP triggers.

CardsFinders Financial Tool Suite

Cloudflare Pages, DNS Architecture, Web Engineering
Live:: Credit Cards India - Compare & Apply | cards.

- Deployed a financial comparison and evaluation web engine on Cloudflare Pages, configuring custom DNS architectures, edge-caching layers, and domain-level email routing.
- Engineered structured data tables and client-side filtering tools to evaluate credit card rewards and financial products.

Experimental Web & Utility Suite (`myuselessweb.com` & Production Micro-Tools)

- Built and deployed interactive web experiments (`myuselessweb.com`) exploring event-driven frontend interactions, randomized API fetching, and edge delivery.

- Developed industrial production micro-utilities including `image-sequence-renamer.pages.dev` to automate asset management for manufacturing teams.

Photovoltaic Sensorless Solar Tracker:

Live:: https://docs.google.com/presentation/d/1WsoxJpkOGv4GCKoAZFYMt03jj_0Yq4AvVkb0T4s2bPY/edit?usp=sharing | https://drive.google.com/file/d/1fcZVVkoQ_b0vqO3GWKNmk0JH5dNND53V/view?usp=sharing

- **Engineered a dual-axis solar tracking algorithm that eliminates external light-dependent resistors (LDRs) and auxiliary optical sensors.**
- **Utilized the intrinsic photovoltaic voltage and current characteristics of the solar panel itself as the diagnostic light-intensity sensor, dynamically calculating maximum power point vectors to actuate motor positioners with minimal component overhead.**

Gas Detection Safety System

Live:: https://wokwi.com/projects/453321379118788609 |

- Built an embedded gas-hazard detection circuit integrating an Arduino Uno with an MQ-2 electrochemical gas sensor.
- Programmed analog-to-digital threshold comparison logic, automatic alarm actuation, and sensor calibration routines to trigger emergency alerts under volatile gas leak scenarios.

Publications & Research Credentials

Lead Author, IEEE Xplore Publication

"Automated Object Detection and Billing Architecture for Smart Retail Systems"
IEEE Xplore, Document ID: 10627276
Live:: Smart Shopping Cart | IEEE Conference Publication | IEEE Xplore

- **Focus:** Real-time edge inference, constrained ARM compute optimization, and embedded computer vision pipelines.